5/2 Valves and Pilot Air

Press once: the blade goes out, then comes back by itself.



National 5 Engineering Science

2018 N5 Engineering Science: scan the code and try these.

Q7

Page 5. Sketch it in your jotter.

Scan for the paper

Marking scheme

2024 N5 Engineering Science: scan the code and try these.

Q1(a)

Page 2. Answer in your jotter.

Scan for the paper

Marking scheme

PART 1

The 5/2 Valve and Pilot Air

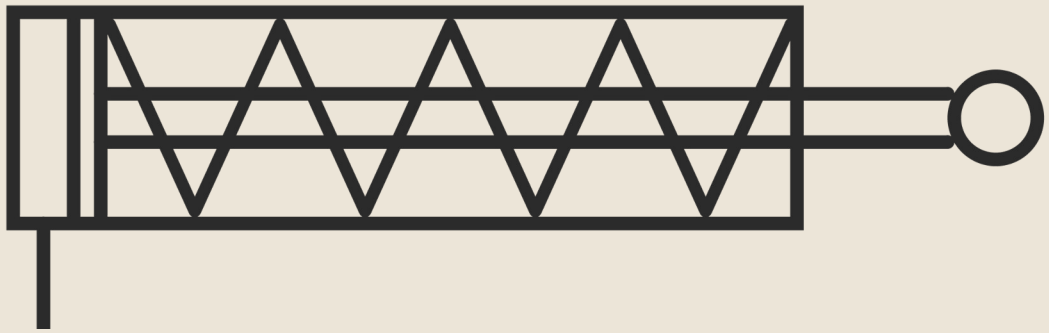
To describe how pilot air and a 5/2 valve control a double-acting cylinder.

By the end of this part… (p.10)

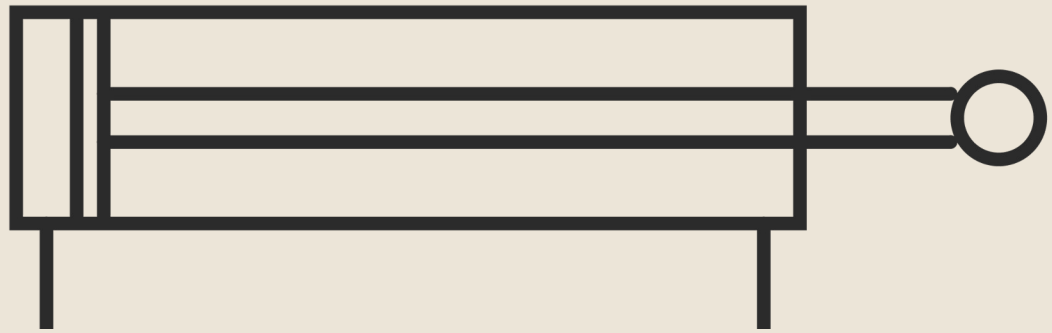


1. I can **describe** how a double-acting cylinder outstrokes and instrokes.
2. I can **read** a 5/2 valve symbol: its ports, its two states, main air and exhausts.
3. I can **describe** how pilot air changes the state of a 5/2 valve.
4. I can **describe** a circuit that outstrokes and instrokes a piston automatically.

Pn2's piston came back on a spring. This one has no spring. What brings it back?



Single-acting: a spring inside



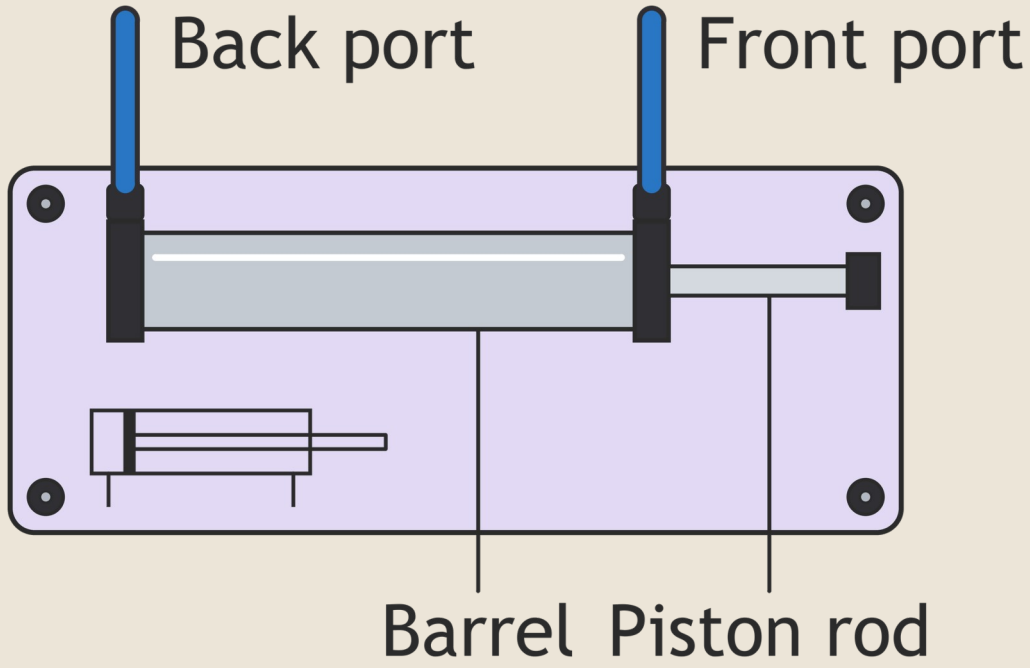
Double-acting: no spring

The Double-Acting Cylinder (p.10)

Main air on the left of the piston makes it **outstroke**.

Main air on the right of the piston makes it **instroke**.

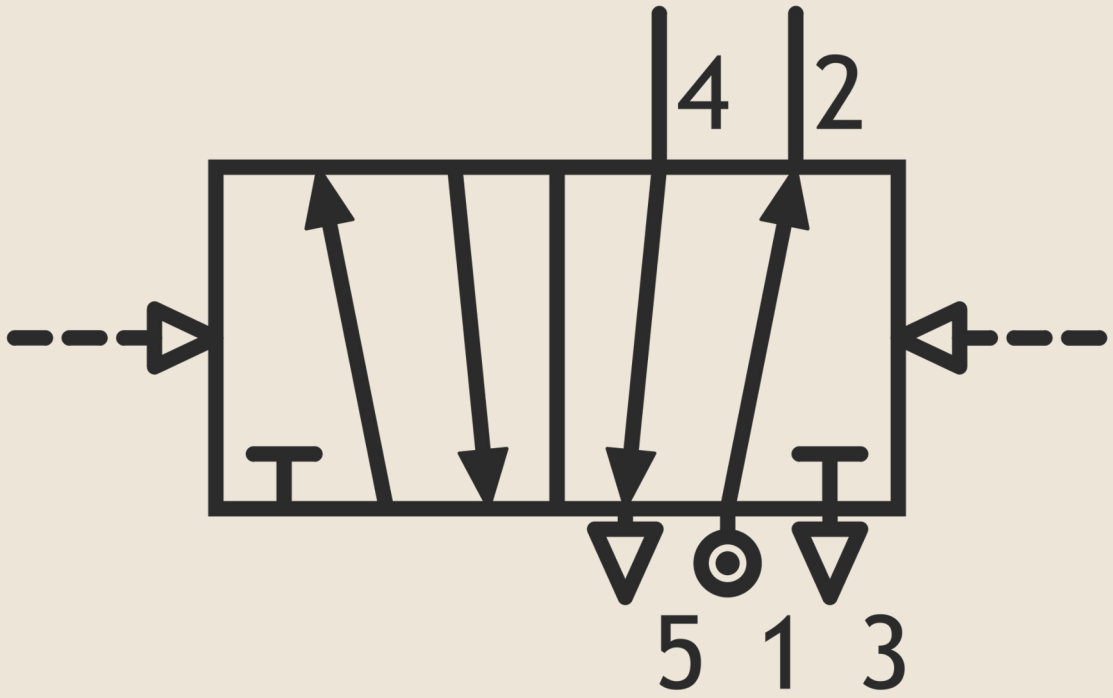
A double-acting cylinder has no spring: it is powered by air in **both** directions.



The 5/2 Valve (p.10)

5/2 means **5** ports and **2** states.

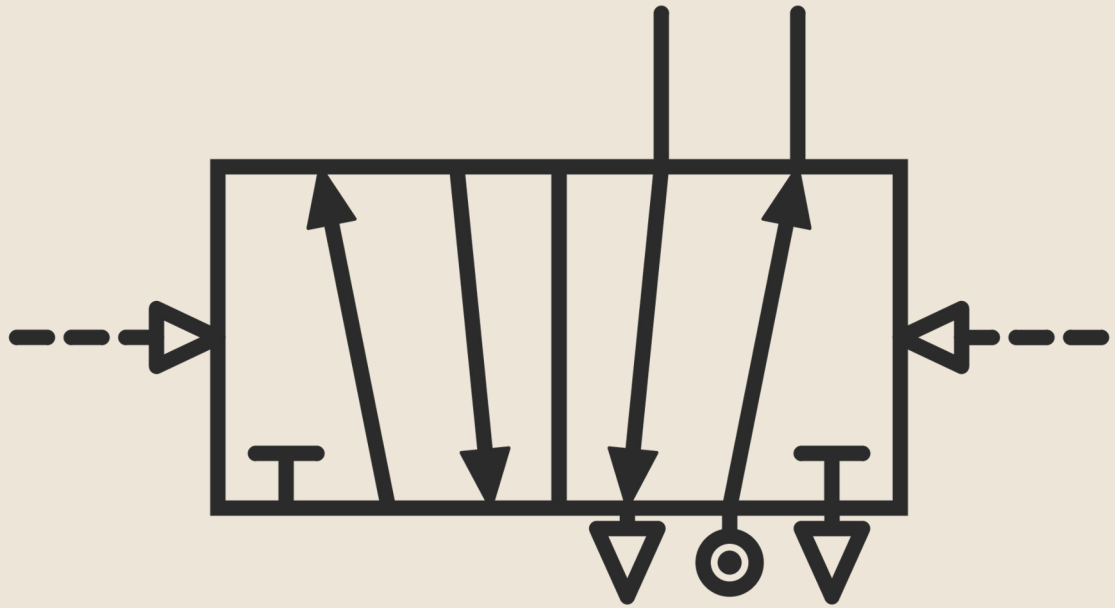
Port 1 is main air. Ports 2 and 4 go to the **cylinder**. Ports 3 and 5 are **exhausts**.



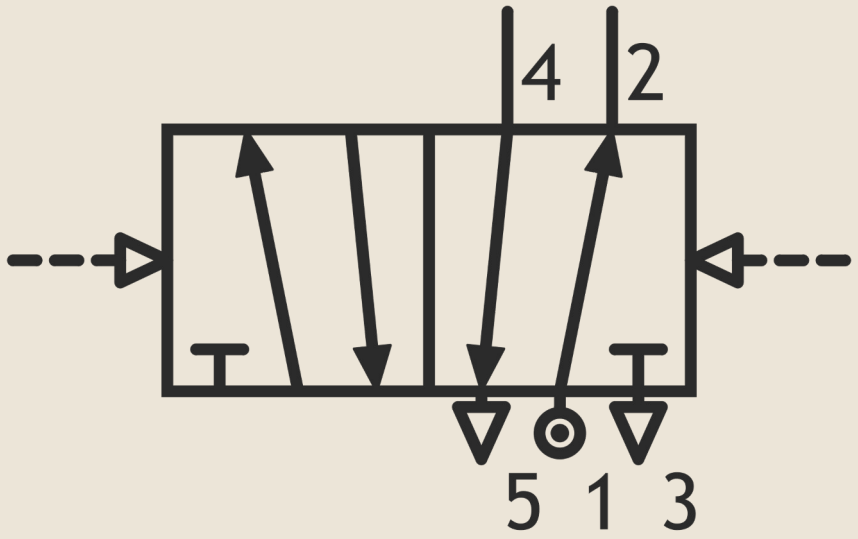
Pilot Air Changes Its State (p.10)

When pilot air reaches one end, the 5/2 valve **changes state**.

The valve stays in that state until pilot air reaches the **other** end.



Show Me: What Does Each Port Do?



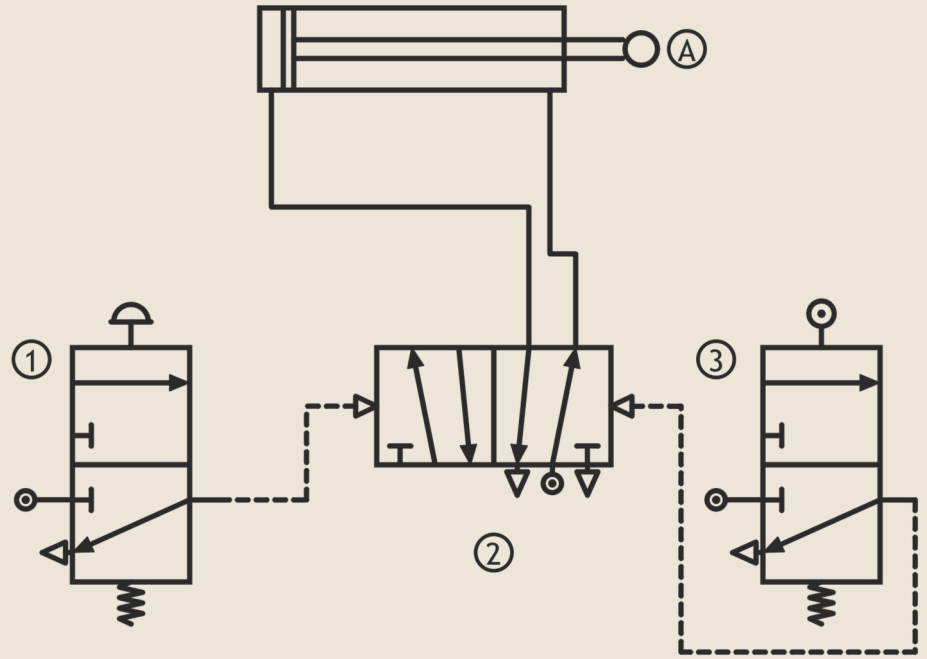
Port 1	Main air in
Ports 2 and 4	To the cylinder
Ports 3 and 5	Exhausts
Pilot air	Changes its state

Reciprocating Motion (p.10)

1. Valve 1 is actuated. Pilot air changes the state of valve 2, so the piston **outstrokes**.

2. At the end of its outstroke the piston **actuates** valve 3.

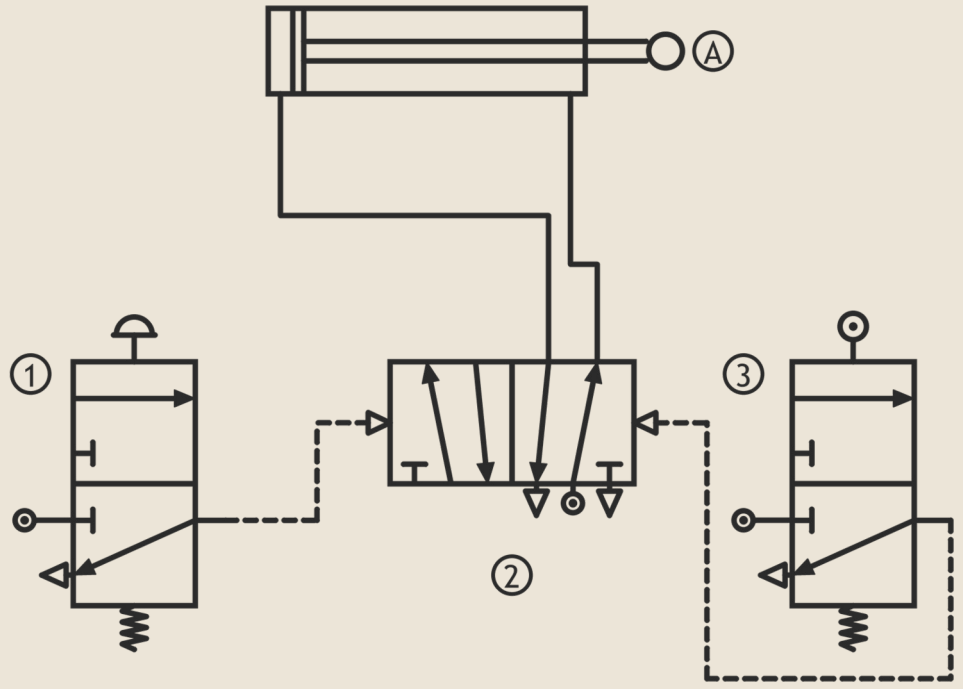
3. Pilot air from valve 3 changes the state of valve 2 back, so the piston **instrokes**.



Words That Get the Marks (p.11)

Each mark needs three things: the **valve** by its number, what it does, and what the **named** piston does.

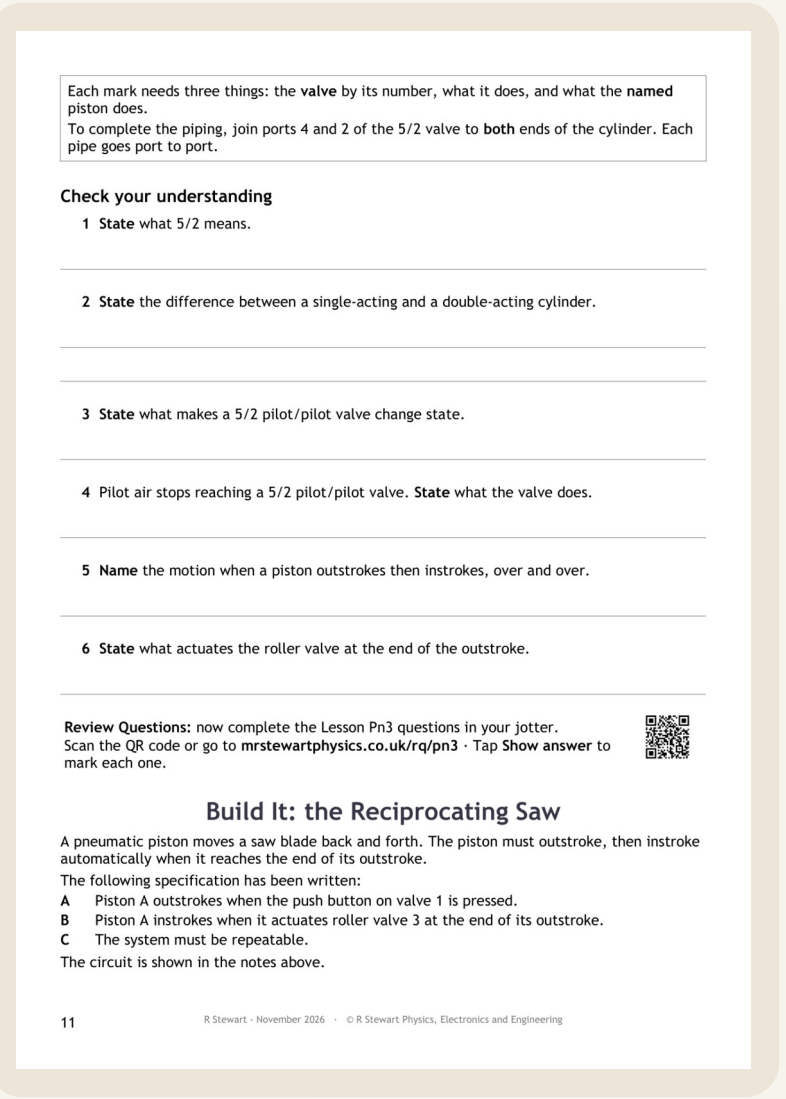
To complete the piping, join ports 4 and 2 of the 5/2 valve to **both** ends of the cylinder. Each pipe goes port to port.



Check Your Understanding (pp 10-11)

In your booklet

Answer the **Quick check** on page **10**.
Answer Questions **1** to **6** on page **11**.



Each mark needs three things: the **valve** by its number, what it does, and what the **named** piston does.
To complete the piping, join ports 4 and 2 of the 5/2 valve to **both** ends of the cylinder. Each pipe goes port to port.

Check your understanding

1 **State** what 5/2 means.

2 **State** the difference between a single-acting and a double-acting cylinder.

3 **State** what makes a 5/2 pilot/pilot valve change state.

4 Pilot air stops reaching a 5/2 pilot/pilot valve. **State** what the valve does.

5 **Name** the motion when a piston outstrokes then instrokes, over and over.

6 **State** what actuates the roller valve at the end of the outstroke.

Review Questions: now complete the Lesson Pn3 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn3** · Tap **Show answer** to mark each one.

Build It: the Reciprocating Saw

A pneumatic piston moves a saw blade back and forth. The piston must outstroke, then instroke automatically when it reaches the end of its outstroke.

The following specification has been written:

A Piston A outstrokes when the push button on valve 1 is pressed.
B Piston A instrokes when it actuates roller valve 3 at the end of its outstroke.
C The system must be repeatable.

The circuit is shown in the notes above.

Answers: Check Your Understanding (pp 10-11)



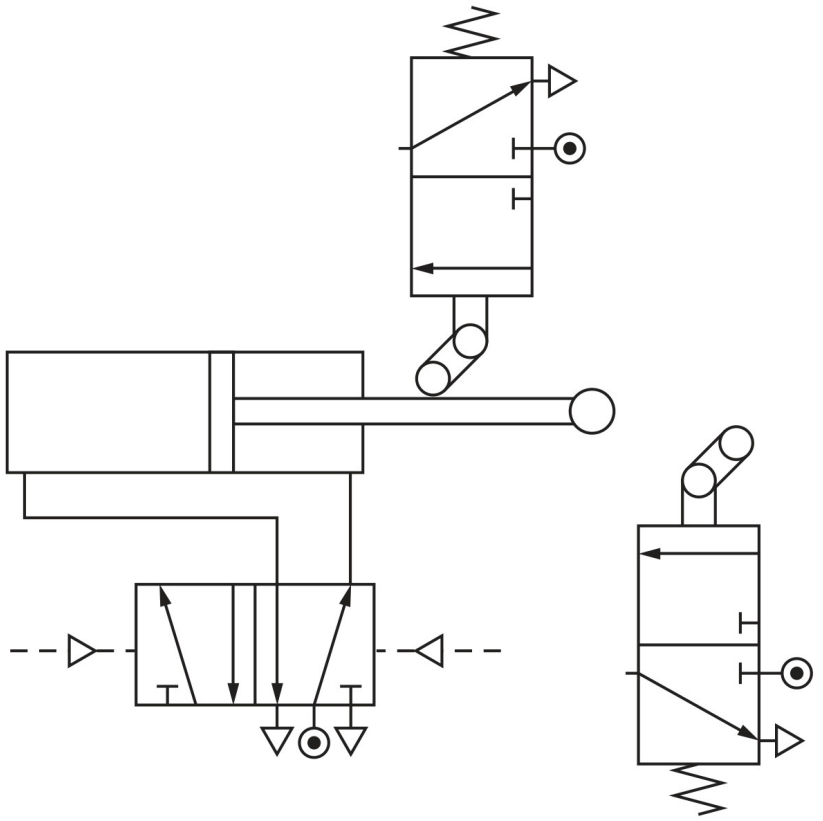
Question	Answer
Quick check p.10	Ports **2** and **4**.
1	**5** ports and **2** states.
2	Single: **spring** back. Double: **air** both ways.
3	**Pilot air** reaching one end.
4	It **stays** in the same state.
5	**Reciprocating**.
6	The moving **piston**.

Exam: 2015 Section 1 Question 6

MARKS

6. An incomplete pneumatic circuit to produce reciprocating motion is shown in the diagram below.

Complete the diagram by inserting the missing piping. 2



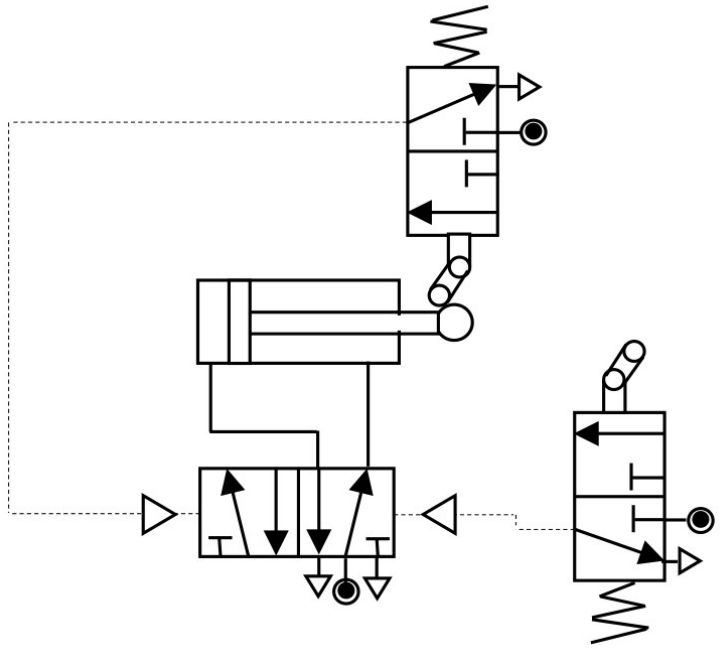
Question 6

2 marks

Page 6 of the paper.

Sketch it in your jotter and add the pipes.

Marking: 2015 Section 1 Question 6

6.			 1 mark for piping (Port to port) x2	2	Do not penalise for no pilot lines or extra pipes. 1 mark where 3/2 valves are piped to the incorrect 5/2 pilot ports.

What Candidates Get Wrong

Pipe the 5/2

ports **4** and **2**, one to **each end** (2026)

Port to port

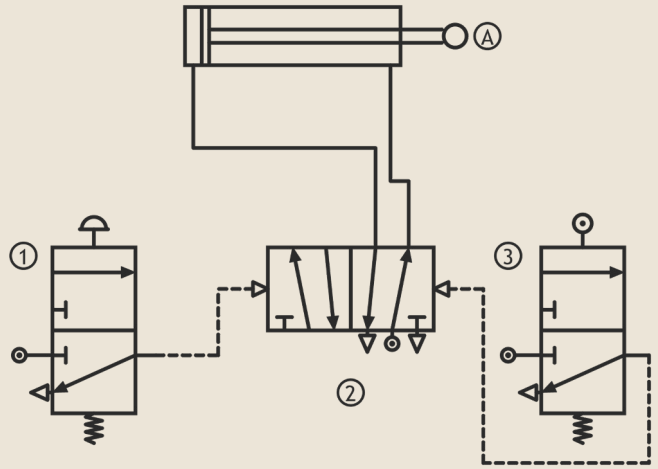
every pipe (2022 MI)

Pilot air

a **dashed** line into the 5/2 (2022 MI)

Say it right

the 5/2 **changes state** (2024, 2025)



Review Questions (p.11)

Lesson Pn3 (online)

Answer the Lesson **Pn3** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn3**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this part… (p.10)



1. I can **describe** how a double-acting cylinder outstrokes and instrokes.
2. I can **read** a 5/2 valve symbol: its ports, its two states, main air and exhausts.
3. I can **describe** how pilot air changes the state of a 5/2 valve.
4. I can **describe** a circuit that outstrokes and instrokes a piston automatically.

2024 N5 Engineering Science: scan the code and try these.

Q15(d)

Page 30. Answer in your jotter.

Scan for the paper

Marking scheme

PART 2

Build It: the Reciprocating Saw

To plan, simulate, build and evaluate a pneumatic circuit, as the assignment asks.

By the end of this part… (pp 11-12)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

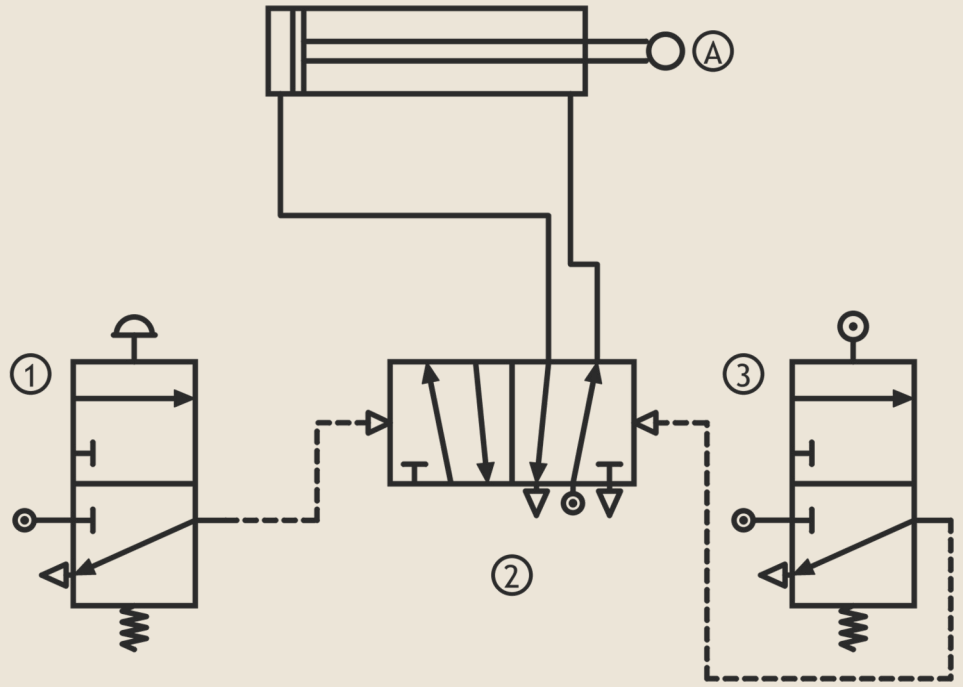
Build It: the Reciprocating Saw (p.11)

A pneumatic piston moves a saw blade back and forth.

A Piston A **outstrokes** when the push button on valve 1 is pressed.

B Piston A **instrokes** when it actuates roller valve 3 at the end of its outstroke.

C The system must be **repeatable**.



Write a Test Plan (p.12)

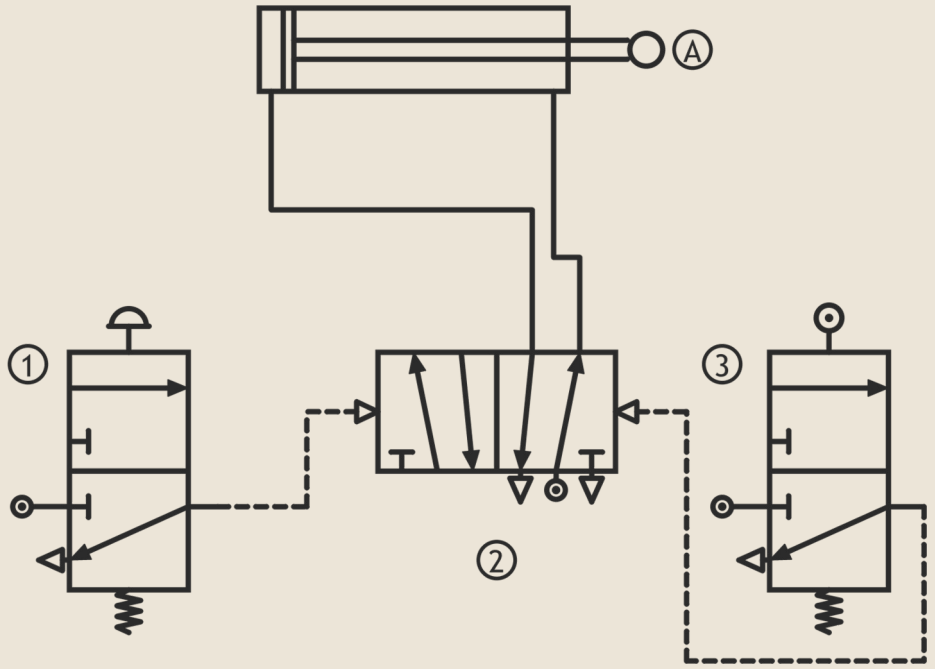


Planned test	**Expected result**	**Test result**
Press the push button on valve 1.	Piston A **outstrokes**.	
Watch piston A reach the end of its outstroke.	Piston A **actuates** valve 3 and **instrokes**.	
Repeat tests 1 and 2 three times.	The same **every time**.	

Simulate It (p.12)

In your booklet

1. **Open** NoPressureSim. **Build** the circuit, as on page 8.
2. **Predict** what happens. **Test** it.
3. Air off: **build** it on the kit. Air on: **test**.

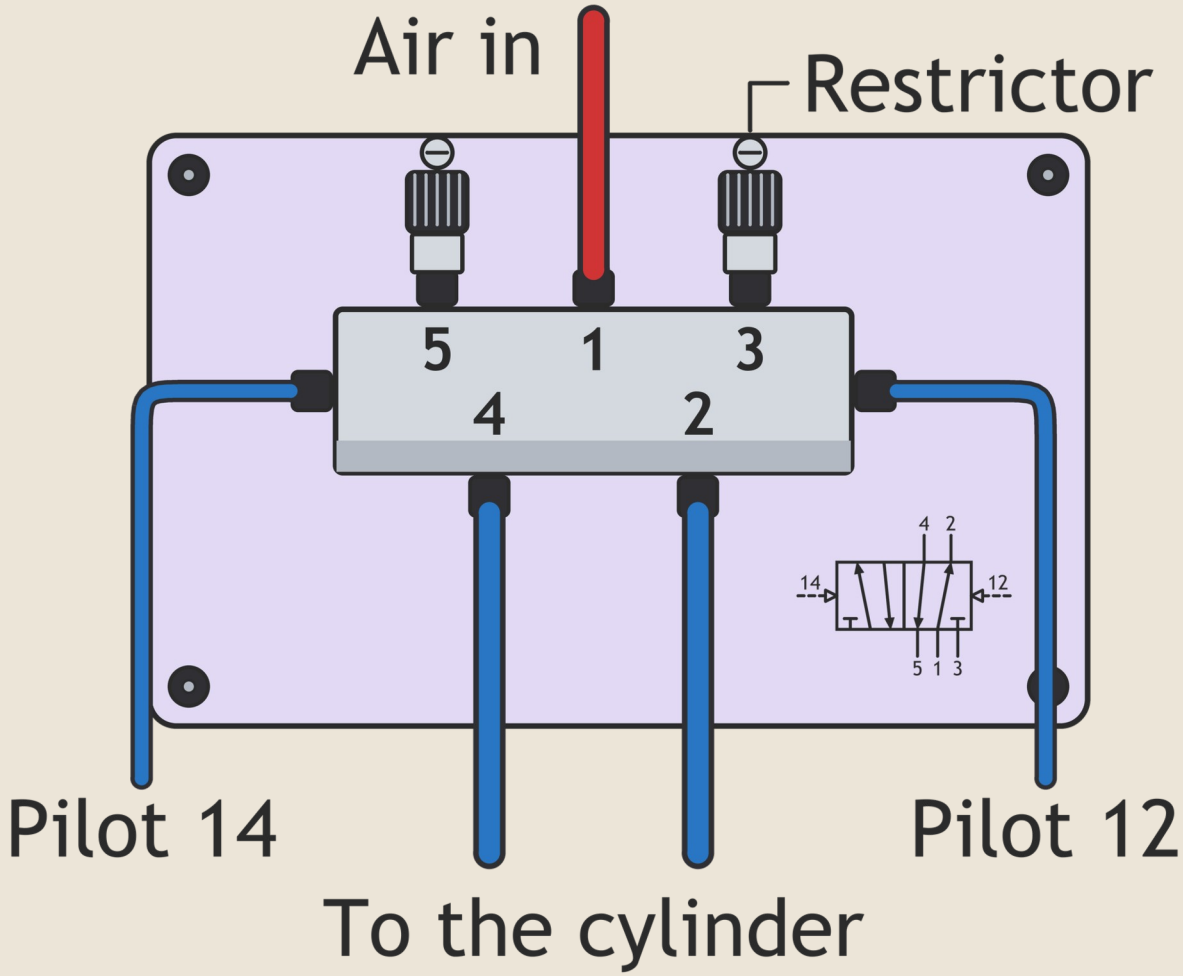


Scan me
NoPressureSim

What to Collect

Collect one set per pair.

1. 3/2 push-button valve
2. 5/2 pilot/pilot valve
3. 3/2 roller valve
4. Double-acting cylinder
5. Seven blue tubes
6. Goggles each



Build It on the Kit (p.12)

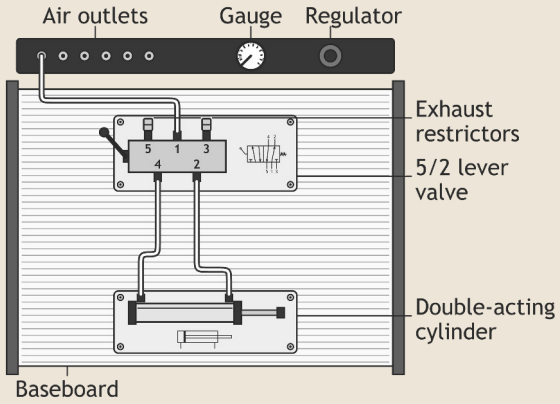
On the kit

Build it, air **off**.

Main air to port **1** of each valve.

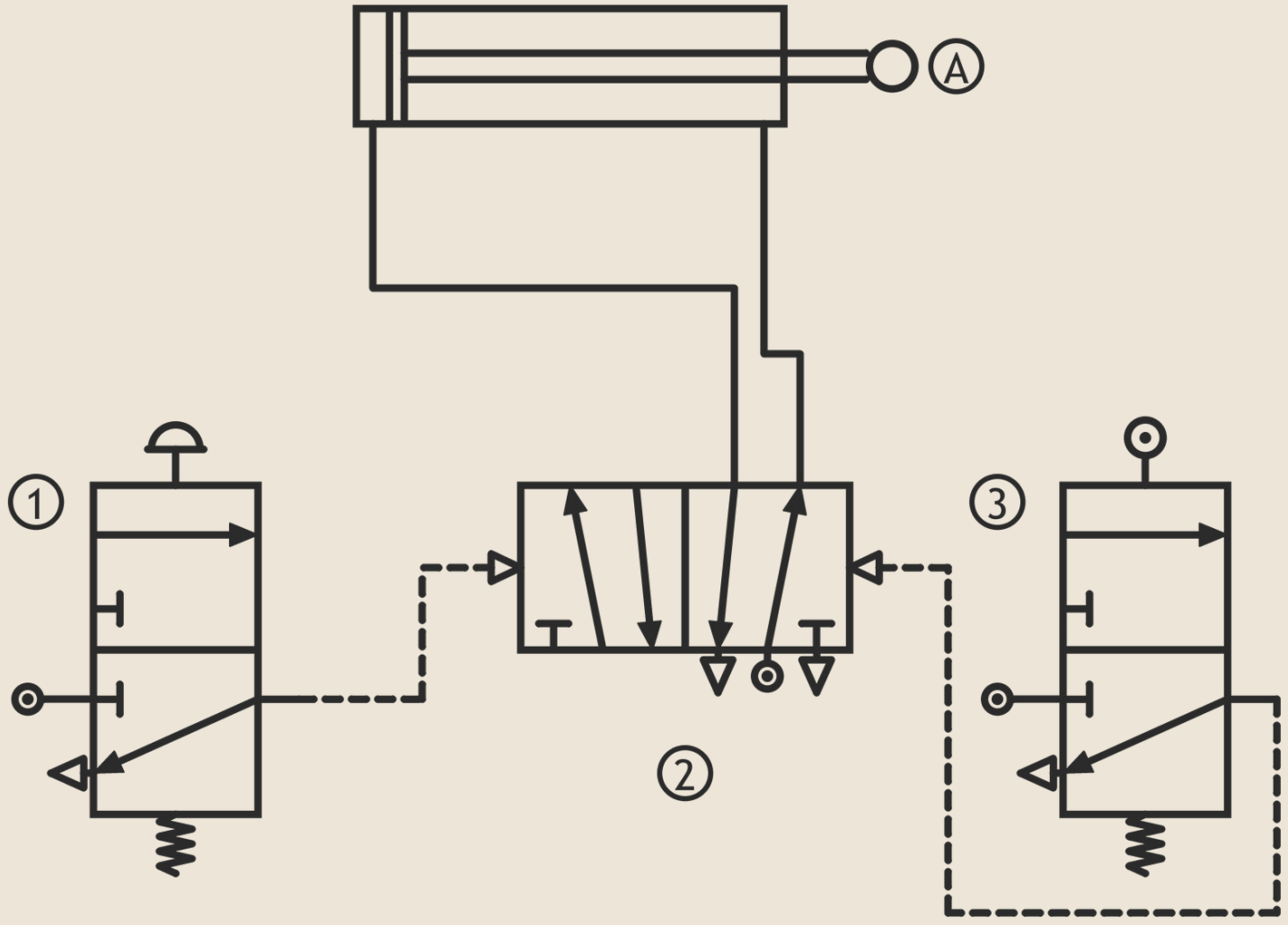
Valve 2: ports **4** and **2** to the cylinder.

Air on. **Test** and **record**.



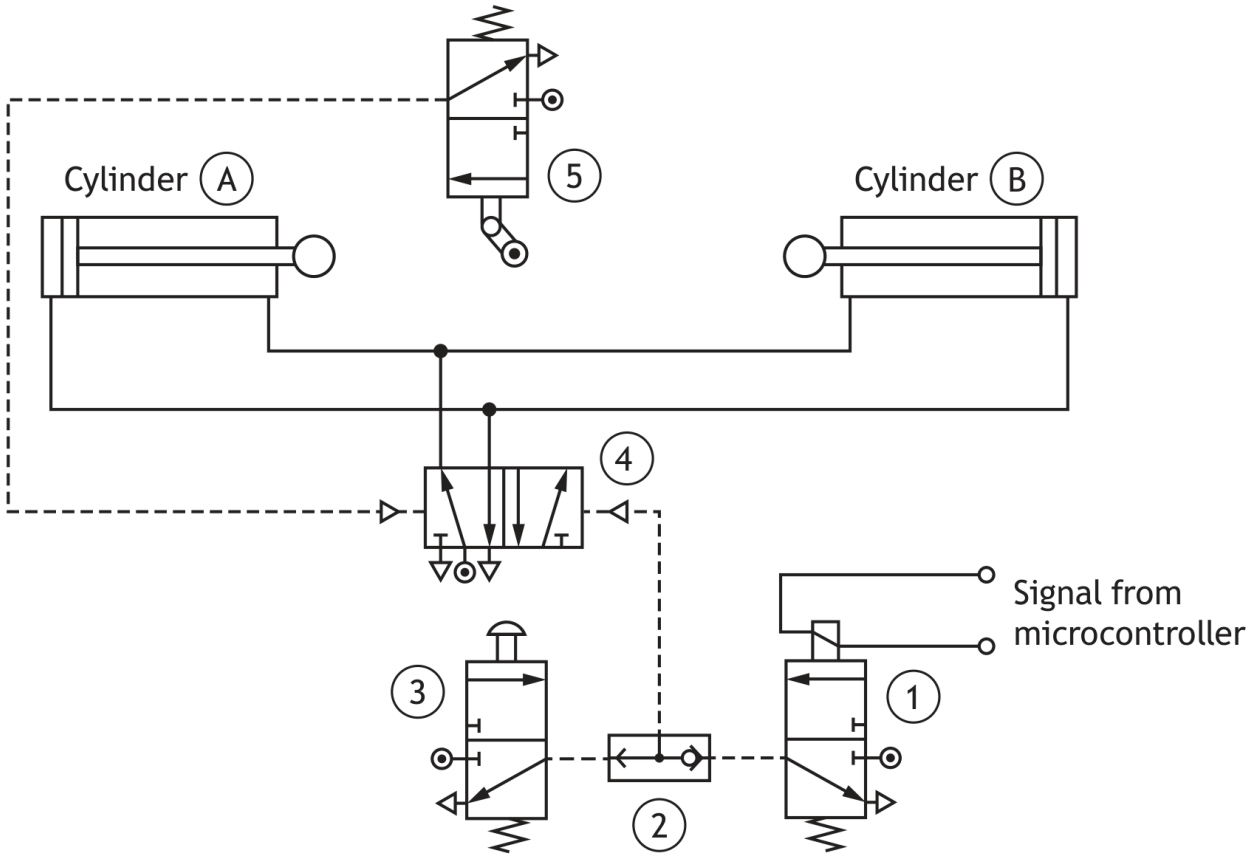
SAFETY 0 bar to build. 1 to 2 bar to test.

Build It: Bench Card 1



Exam: 2017 Section 2 Question 11 (a)

11. A pneumatic circuit used to process components on a production line is shown below.



(a) Describe, using appropriate terminology, the operation of the circuit. **3**

Question 11 (a)

3 marks

Page 14 of the paper.

Valve **2** lets valve 1 **OR** valve 3 send the pilot air.

Marking: 2017 Section 2 Question 11 (a)

11.	(a)		Valve 1 OR 3 is actuated. Air is sent to valve 4 causing both pistons to outstroke. The piston from cylinder A will actuate valve 5 causing both pistons to instroke.	**3**	1 mark for OR statement 1 mark for both pistons outstroking 1 mark for valve 5 **and** both pistons instroking

How It's Marked: One Mark per Statement

1 mark

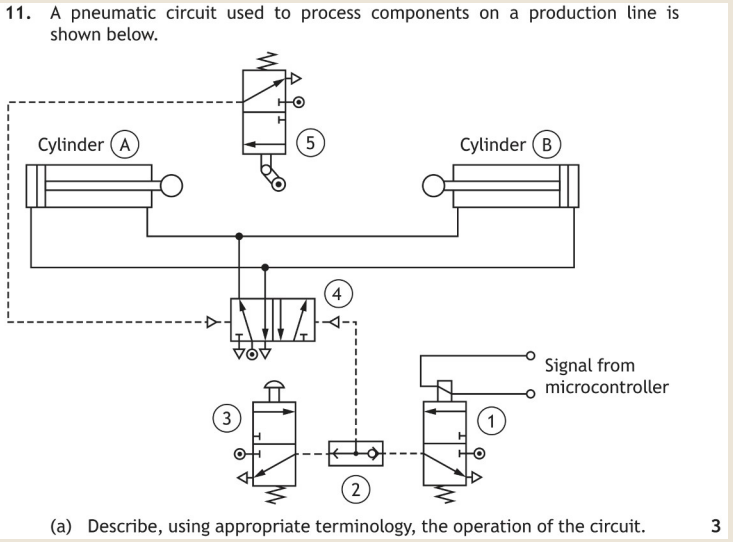
Valve **1 OR 3** is actuated.

1 mark

Air is sent to valve **4** causing **both pistons** to **outstroke**.

1 mark

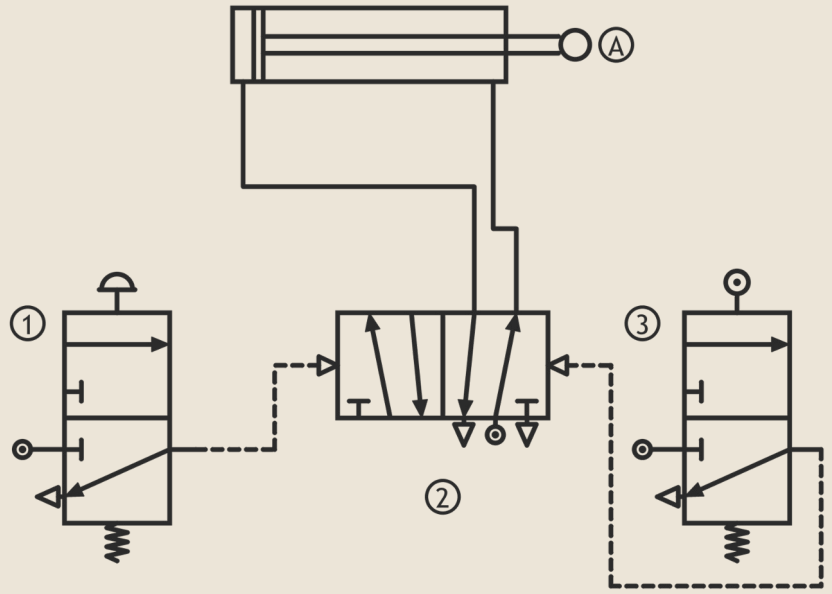
The piston from cylinder A will **actuate valve 5** causing both pistons to **instroke**.



Evaluate (p.12)

In your booklet

For each specification point, **state** if it was met.
Explain why, using your **test results**.
Say what you did **and** what the piston did.



Improve and Justify (p.12)

In your booklet

The operator has to stand **next to the saw blade** to press the button.
Describe a modification that would solve this.
Justify it: say **why** it solves the problem.



Answers: Evaluate (p.12)

Point	Met?	Explanation
A	Yes	**In test 1**, the button was **pressed**, pilot air **changed the state** of valve 2, **so** piston A **outstroked.**
B	Yes	**In test 2**, piston A **actuated** valve 3, valve 2 **changed state** back, **so** piston A **instroked.**
C	Yes	**In test 3**, piston A did the same **every time**, over three repeats.

Answers: Improve and Justify (p.12)

	Answer
Modification	Fit a **guard** over the saw blade.
Justification	The guard keeps the operator's hands **away from the blade**, **so** they cannot be cut while it moves.

What Candidates Get Wrong

Right words

the 5/2 **changes state** (2024, 2025)

One mark each

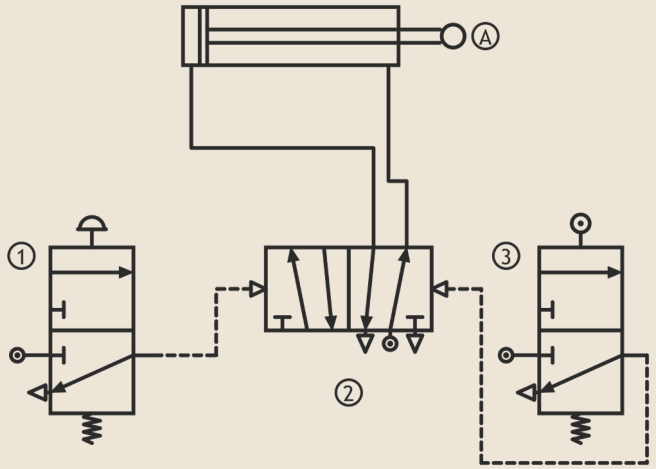
a **valve** and a **piston** (2017 MI)

Evaluate

In test 1 … so … (2026)

Air supply

on **every** valve (2025)



Look back — by the end of this part… (pp 11-12)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

Scan: page 6

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 3)

Title: Photo: Korrupt, CC BY 4.0, via Wikimedia Commons

Starter: 2018 N5 Engineering Science, © Qualifications Scotland (SQA). 2018 question paper and marking instructions; 2024 N5 Engineering Science, © Qualifications Scotland (SQA). 2024 question paper and marking instructions

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Exam: 2015 Section 1 Question 6: © Qualifications Scotland (SQA). 2015 National 5 Engineering Science Section 1, Q6

Marking: 2015 Section 1 Question 6: © Qualifications Scotland (SQA). 2015 National 5 Engineering Science Section 1, Q6 marking instructions

Credits (2 of 3)

What Candidates Get Wrong: © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2024, 2025 and 2026; 2022 S2 Q15(a) marking instructions; © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2024, 2025 and 2026; 2017 S2 Q11(a) marking instructions

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

Exam: 2017 Section 2 Question 11 (a): © Qualifications Scotland (SQA). 2017 National 5 Engineering Science Section 2, Q11

Marking: 2017 Section 2 Question 11 (a): © Qualifications Scotland (SQA). 2017 National 5 Engineering Science Section 2, Q11 marking instructions

How It's Marked: © Qualifications Scotland (SQA). 2017 National 5 Engineering Science Section 2, Q11 marking instructions

Credits (3 of 3)

Course specification: © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.6

R Stewart - November 2026